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Glabretal-type triterpenoid from the root bark of *Dictamnus dasycarpus* ameliorates collagen-induced arthritis by inhibiting Erk-dependent lymphocyte proliferation

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ABSTRACT

Ethnopharmacological relevance: The root bark of *Dictamnus dasycarpus* Turcz. (Rutaceae) has been used as a traditional herbal medicine to treat various inflammatory diseases in East Asia. We have showed previously that a glabretal type triterpenoid (dictabretol A) from *D. dasycarpus* root bark has immunosuppressive activity.
Aim of the study: This study was conducted to define the molecular mechanism of how dictabretol A inhibits lymphocyte proliferation and to evaluate the therapeutic efficacy of dictabretol A in an animal model of rheumatoid arthritis.
Materials and methods: Various murine immune cells (T cells, B cells, and macrophages) and splenocytes were used to study the anti-proliferative effect of dictabretol A *in vitro*. A collagen-induced arthritis model was also used to examine the therapeutic effect of dictabretol A *in vivo*.
Results: Dictabretol A specifically inhibited lymphocyte proliferation by blocking the cell cycle transition from the G_1 to the S phase. This effect was achieved by blocking Erk1/2, nuclear factor kappa B, and the C-myc axis of cell cycle progression. Further dictabretol A treatment alleviated the severity of collagen-induced arthritis.
Conclusion: Our results reveal the molecular mechanism for the anti-lymphoproliferative effect of dictabretol A and show the therapeutic efficacy of dictabretol A for rheumatoid arthritis.
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1. Introduction

Immunity is a complex and sophisticated network that orchestrates various subsets of immune cells. Both innate and adaptive immunity interact to protect the body against threats from foreign antigens, which are mainly associated with microbial pathogens. However, an excess immune response or activation of lymphocytes by self-antigens causes hypersensitivity or autoimmune disease that destroys specific tissues or cells (Ganguly et al., 2013; Wu and Zarrin, 2014). Also, inappropriate immune responses cause graft rejection when a MHC mismatched organ is transplanted to a recipient (Wood and Goto, 2012). Therefore, immunosuppressive agents are necessary to control immune homeostasis in patients with immune-related diseases.

The root bark of *Dictamnus dasycarpus* Turcz. (Rutaceae) have been widely used as an herbal medicine for the treatment of cough, jaundice, rheumatism, and skin inflammation in East Asia (Chang et al., 2001; Lei et al., 2008; Wu, 2005). We showed previously that a glabretal type triterpenoid (dictabretol A) from this plant has anti-proliferative activity against immune cells (Kim et al., 2015). In this study, we defined the anti-lymphoproliferative mechanism of dictabretol A and investigated the effect of dictabretol A on disease progression in a collagen-induced arthritis (CIA) model to evaluate the potency of dictabretol A as a novel immunosuppressive agent.

Abbreviations: CFSE, carboxyfluorescein succinimidyl ester; ChIP, chromatin immunoprecipitation; CIA, collagen induced arthritis; IRE, internal regulatory element; OVA, ovalbumin peptide; PI, propidium iodide; URE, upstream regulatory element

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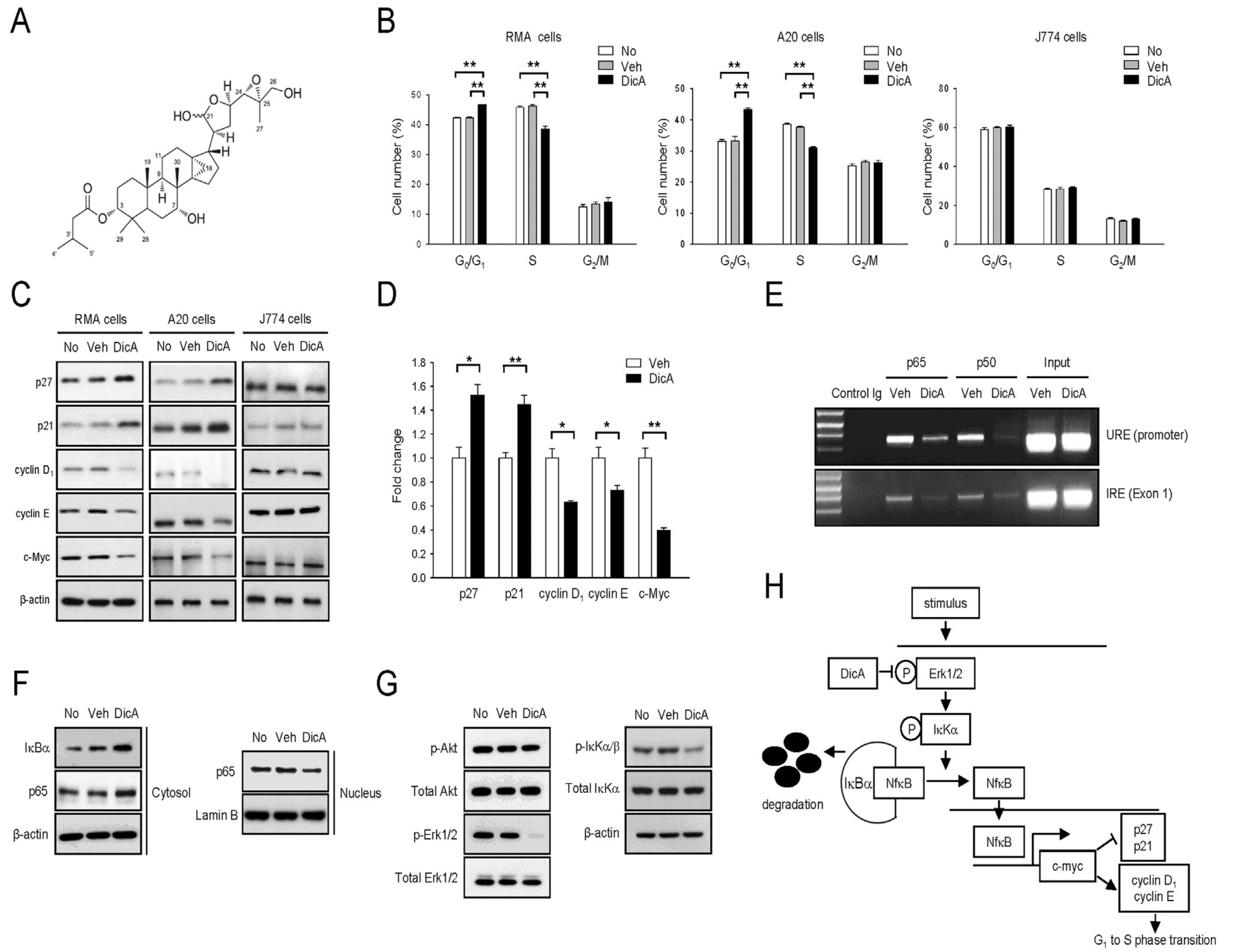

Fig. 1. Dictabretol A specifically inhibits lymphocyte proliferation by blocking the Erk1/2, NFκB, and C-myc axis of the cell cycle transition from the G1 phase to the S phase. (A) Chemical structure of dictabretol A. (B–C) Indicated cells (10^6 cells/ml) were seeded in 6-well plates and cultured with dictabretol A (2 μM) or 0.1% DMSO (vehicle) for 24 h. Then, the cell cycle was analyzed by PI staining (B) and protein expression was examined by immunoblotting (C). No, untreated cells; Veh, vehicle (0.1% DMSO) treated cells; DicA, dictabretol A treated cells. Bars represent mean ± SEM of three independent experiments. *$p < 0.05$, **$p < 0.01$. (D–G) After the 24 h treatment, RMA cells were analyzed for mRNA expression by quantitative real-time PCR (D), NFκB binding activity on the regulatory regions of c-myc gene by ChIP assay (E), the nuclear translocation of p65 by immoblotting (F), phosphorylation of proteins by immoblotting (G). No, untreated cells; Veh, vehicle (0.1% DMSO) treated cells; DicA, dictabretol A treated cells; Control Ig, isotype control for antibody; Input, genomic DNA (positive control for PCR); p-Akt, phosphorylated Akt, p-Erk, phosphorylated Erk, p-IκK, phosphorylated IκK. Bars represent mean ± SEM from three independent experiments. *$p < 0.05$, **$p < 0.01$. (H) Schematic diagram depicting the molecular mechanism of how dictabretol A (DicA) inhibits cell cycle transition from the G1 phase to the S phase. Ⓟ, phosphorylated protein.

2. Materials and methods

2.1. Antibodies and reagents

See the Materials and Methods section of Supplementary Material for the details on the antibodies and reagents.

2.2. Dictabretol A purification

The dried root bark of *D. dasycarpus* was purchased from Kyung Dong market, Seoul, Korea, in January 2008, and was authenticated by Emeritus Professor Kyong Soon Lee, College of Pharmacy, Chungbuk National University, Cheongju, Korea. A voucher specimen (no. SS45-01142008) was deposited at the College of Life Sciences and Biotechnology, Korea University, Seoul, Korea. Dictabretol A was purified from the plant as described previously (Kim et al., 2015). See the Materials and Methods section of the Supplementary Material for details on dictabretol A purification.

2.3. Cell culture

A20 cells (mouse B cells), J774 cells (mouse macrophages), and RMA cells (mouse T cells) were obtained from the American Type Culture Collection (Rockville, MD, USA). See the Materials and Methods section of the Supplementary Material for details on cell culture. The carboxyfluorescein succinimidyl ester (CFSE) labeling assays, cell cycle analyses using propidium iodide (PI) staining, chromatin immunoprecipitation (ChIP) assays, flow cytometry analyses, quantitative real-time polymerase chain reaction (PCR) analyses, and western blot analyses were performed as described previously (Lee et al., 2012, 2013; Cho et al., 2013). Primer sequences used for quantitative real-time PCR analyses and ChIP assays are listed in Supplementary Tables 1 and 2.

2.4. Toxicity test of Dictabretol A

See the Materials and Methods section of the Supplementary Material for details on the toxicity test of *Dictabretol A*.

2.5. Animal experiments

See the Materials and Methods section of the Supplementary Material for details on the CIA model. Enzyme-linked immunosorbent assays (ELISAs) were performed as described previously to determine serum levels of collagen-specific IgG_1 and IgG_{2a} (Kleinau et al., 2000). Isotype levels of IgG_1 and IgG_{2a} were measured as arbitrary units using pooled sera from collagen-hyperimmunized mice as the standard. Serum IL-1β and TNF-α levels were measured using ELISA kits (Invitrogen, Carlsbad, CA, USA).

2.6. Statistical analysis

The mean values among more than three experimental groups were compared by ANOVA analysis and comparison between two experimental groups was analyzed by Student's *t*-test using IBM SPSS statistics software, ver. 21.0 (SPSS Inc., Chicago, IL, USA). Significant differences at a confidence level of 95% and 99% are labeled with asterisks (*) and (**), respectively, on each graph.

3. Results and discussion

3.1. Dictabretol A induces cell cycle arrest from the G_1 phase to S phase transition in lymphocytes through Erk1/2 pathway

The molecular formula of dictabretol A is $C_{35}H_{56}O_7$ with eight degrees of unsaturation (Fig. 1A). No significant cell death was observed in splenocytes and cell lines (RMA, A20 and J774 cells) treated with up to 2 μM of dictabretol A (Supplementary Fig. S1). Next, we performed CFSE labeling assay as an *in vitro* model system for cell proliferation, showing that dictabretol A (2 μM) treatment effectively inhibits both T and B cell proliferation when compared to those of untreated cells or 0.1% DMSO (vehicle) treated cells (Supplementary Fig. S2). However, dictabretol A did not inhibit proliferation of macrophages (J774 cells) (Supplementary Fig. S2). It has been demonstrated that $CD4^+$ T cells (helper T cells) are activated by ovalbumin peptide (OVA) administration in the spleen of OT-II mice (Barnden et al., 1998). To confirm these results using primary cells, OVA-stimulated OT-II splenocytes were labeled with CFSE and treated with 0.5 μM dictabretol A. Dictabretol A-treated $CD4^+$ T cells showed less proliferation activity when compared to that of untreated and vehicle-treated cells (Supplementary Fig. S3). Then, we performed the cell cycle analysis using PI staining in the absence or presence of dictabretol A administration. Lymphocytes (RMA and A20 cells) treated with dictabretol A were arrested at the G_1 to S phase transition, but dictabretol A had no effect on cell cycle progression in macrophages (J774 cells) (Fig. 1B). These results indicate that dictabretol A specifically inhibits lymphocyte proliferation through cell cycle arrest at the G_1 to S phase transition.

To understand the potential molecular mechanisms underlying the anti-proliferative effects, specifically in G_1 to S phase transition, of dictabretol A, we evaluated the signaling events induced by dictabretol A in lymphocyte using immunoblotting and quantitative real-time PCR. The results clearly indicated that dictabretol A increased mRNA transcripts of tumor suppressor genes (p27 and p21) and down-regulated mRNA transcripts of c-myc and cyclins (cyclin D_1 and cyclin E) in lymphocytes (Fig. 1C and D). It has been demonstrated that c-myc inhibits transcription of tumor suppressor genes (p27 and p21) and induces transcription of cyclins (cyclin D_1 and cyclin E) (Facchini and Penn, 1998). In addition, c-myc transcription is initiated by NFκB binding activity on the regulatory regions of the c-myc gene (Lee et al., 1995). Therefore, we measured NFκB binding activity on the c-myc gene using a Chip assay. The upstream regulatory element (URE) in the promoter and the internal regulatory element (IRE) in exon 1 within the c-myc gene bind with the p65 and p50 subunits of NFκB (Lee et al., 1995). As shown in Fig. 1E, binding of both NFκB subunits to the URE and IRE was severely impaired after dictabretol A treatment. To confirm these results, proteins from the cytosolic or nuclear fraction of dictabretol A-treated cells were purified, and the NFκB (p65) and IκBα protein expression patterns were analyzed by immunoblotting. Consistent with the Fig. 1E result, cytosolic IκBα protein expression increased and nuclear p65 protein expression decreased in dictabretol A-treated cells, compared with those of untreated or vehicle-treated cells (Fig. 1F).

Since Akt or Erk signaling pathway mediates cell cycle progression from the G_1 to S phase transition in response to an extracellular stimulus (Torii et al., 2006), we further measured Akt and Erk phosphorylation by immunoblotting. Interestingly, the Erk1/2 phosphorylation decreased but that of Akt did not change after treating RMA cells with dictabretol A (Fig. 1G). We also measured IκKα phosphorylation, as the downstream target molecule of the Erk1/2 pathway is IκKα (Zhao and Lee, 1999). The results indicated a decrease in the IκKα phosphorylation level after treating RMA cells with dictabretol A (Fig. 1G). Taken together, the combined results strongly suggest that dictabretol A induces cell cycle arrest from the G_1 phase to the S phase transition by inhibiting Erk1/2, NFκB, and the C-myc axis of cell cycle progression (Fig. 1H).

3.2. Therapeutic effects of dictabretol A in the CIA model

During chronic joint inflammation, such as rheumatoid arthritis, various subsets of T cells are involved in disease progression and B cells play multiple pathological roles in antibody-dependent and -independent ways (Burmester et al., 2014). Thus, inhibiting lymphocyte proliferation may alleviate the symptoms of arthritis. We investigated the therapeutic effect of dictabretol A in the CIA model. Before assessing therapeutic effect of dictabretol A, we evaluate the *in vivo* toxicity of dictabretol A. Compared with Dictabretol A-untreated group (0 mg/kg), no changes on mortality, clinical signs and body weight in Dictabretol A-treated group (10 mg/kg) (data not shown). These results indicated that no toxicity was found *in vivo* at the dose of 10 mg/kg. We found that the arthritis score and cartilage damage in knee joints decreased significantly in the dictabretol A-treated group compared to those in the vehicle-treated group (Fig. 2A and B). Circulating anti-collagen antibodies and proinflammatory cytokines, such as IL-1β and TNF-α, are major molecular markers to determine the severity or disease progression of CIA (Burmester et al., 2014). We found that serum levels of IL-1β, TNF-α, and collagen-specific antibodies (IgG_1 and IgG_{2a}) decreased significantly in the dictabretol A-treated group compared to those in the vehicle-treated group (Fig. 2C and D).

4. Conclusion

This study defined the molecular mechanism of how dictabretol A has anti-lymphoproliferative effects and demonstrated the therapeutic efficacy of dictabretol A in a rheumatoid arthritis model. Dictabretol A specifically blocked the cell cycle transition from the G_1 phase to the S phase by inhibiting Erk-dependent signaling and may be a candidate novel therapeutic agent to treat chronic inflammatory disease.

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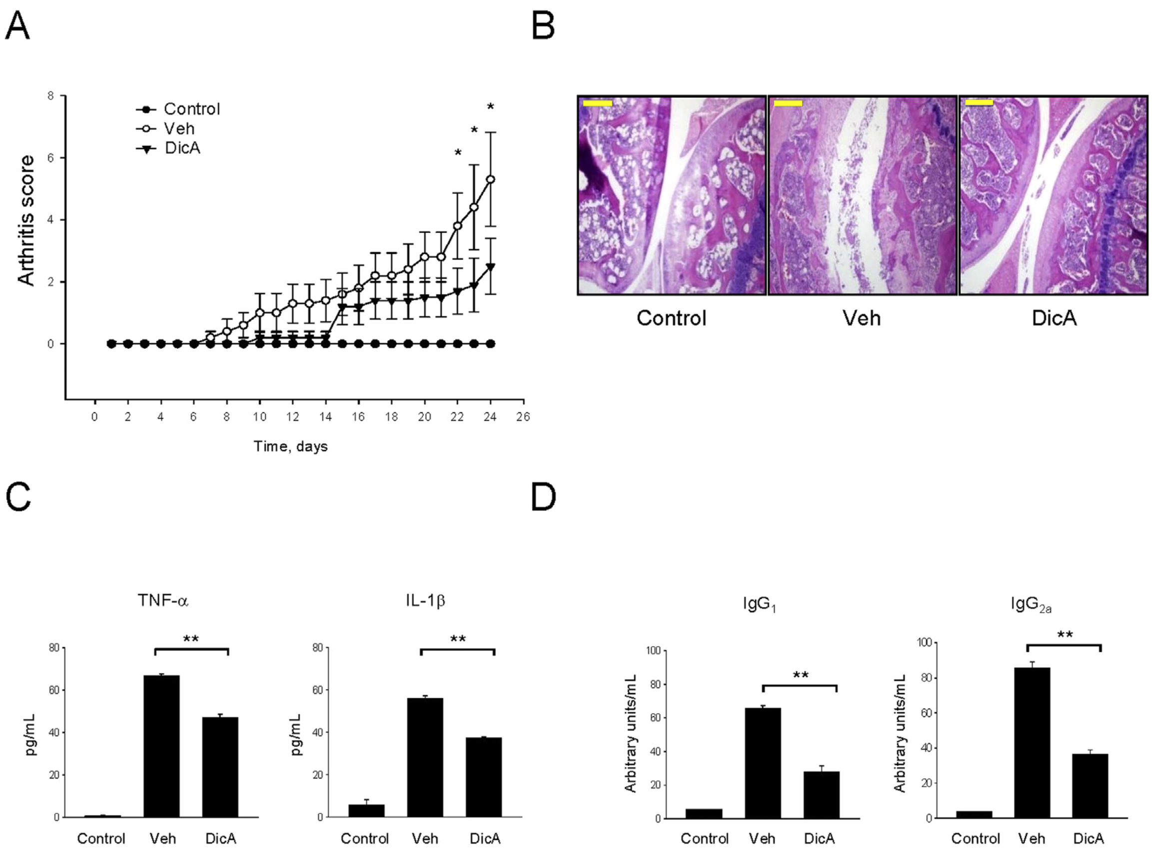

Fig. 2. Therapeutic effect of dictabretol A in the CIA model. Eight week old DBA1/J mice were immunized with bovine type II collagen mixed with adjuvant and injected with 10 mg/kg dictabretol A intraperitoneally once every 2 days for 3 weeks. (A) Arthritis score of mice within each treated group ($n=5$). (B) Histological analysis of the knee joint by H&E staining. (C) Serum TNF-α and IL-1β levels in mice were measured by ELISA. (D) Collagen-specific IgG_1 and IgG_{2a} in sera were measured by ELISA. Control, mice not induced with CIA; Veh, mice induced with CIA and vehicle (0.1% DMSO); DicA, mice induced with CIA and treated with dictabretol A. Bars represent mean ± SEM from three independent experiments. $^{*}p<0.05$, $^{**}p<0.01$.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.jep.2015.10.043.

References

Barnden, M.J., Allison, J., Heath, W.R., Carbone, F.R., 1998. Defective TCR expression in transgenic mice constructed using cDNA-based &agr;-and &bgr;-chain genes under the control of heterologous regulatory elements. Immunol. Cell. Biol. 76, 34–40.
Burmester, G.R., Feist, E., Dörner, T., 2014. Emerging cell and cytokine targets in rheumatoid arthritis. Nat. Rev. Rheumatol. 10, 77–88.
Chang, J., Xuan, L.J., Xu, Y.M., Zhang, J.S., 2001. Seven new sesquiterpene glycosides from the root bark of *Dictamnus dasycarpus*. J. Nat. Prod. 64, 935–938.
Cho, K.W., Bae, J., Lee, S.J., Chun, T., 2013. Expression pattern and functional role of Phc2 during activation of helper T cells after antigenic stimulation. Vitr. Cell. Dev–. 49, 360–370.
Facchini, L.M., Penn, L.Z., 1998. The molecular role of Myc in growth and transformation: recent discoveries lead to new insights. FASEB J. 12, 633–651.
Ganguly, D., Haak, S., Sisirak, V., Reizis, B., 2013. The role of dendritic cells in autoimmunity. Nat. Rev. Immunol. 13, 566–577.
Kim, N., Cho, K.W., Hong, S.S., Hwang, B.Y., Chun, T., Lee, D., 2015. Antiproliferative glabretal-type triterpenoids from the root bark of *Dictamnus dasycarpus*. Bioorg. Med. Chem. Lett. 25, 621–625.
Kleinau, S., Martinsson, P., Heyman, B., 2000. Induction and suppression of collagen-induced arthritis is dependent on distinct Fcγ receptors. J. Exp. Med. 191, 1611–1616.
Lee, H., Arsura, M., Wu, M., Duyao, M., Buckler, A.J., Sonenshein, G.E., 1995. Role of Rel-related factors in control of c-myc gene transcription in receptor-mediated apoptosis of the murine B cell WEHI 231 line. J. Exp. Med. 181, 1169–1177.
Lee, H.H., Lee, S.J., Kim, S., Jeong, S., Na, M., Lee, D.-M., Cheon, Y.P., Lee, K.H., Choi, I., Chun, T., 2012. Porcine testicular extract inhibits T cell proliferation by blocking cell cycle transition from G1 phase to S phase. Biotechnol. Lett. 34, 1225–1233.
Lee, S.J., Bae, J., Kim, S., Jeong, S., Choi, C.Y., Choi, S.P., Kim, H.S., Jung, W.W., Imm, J.Y., Kim, S.H., Chun, T., 2013. Saponins from soy bean and mung bean inhibit the antigen specific activation of helper T cells by blocking cell cycle progression. Biotechnol. Lett. 35, 165–173.
Lei, J., Yu, J., Yu, H., Liao, Z., 2008. Composition, cytotoxicity and antimicrobial activity of essential oil from *Dictamnus dasycarpus*. Food Chem. 107, 1205–1209.
Torii, S., Yamamoto, T., Tsuchiya, Y., Nishida, E., 2006. ERK MAP kinase in G1 cell cycle progression and cancer. Cancer Sci. 97, 697–702.
Wood, K.J., Goto, R., 2012. Mechanisms of rejection: current perspectives. Transplantation 93, 1–10.
Wu, L.C., Zarrin, A.A., 2014. The production and regulation of IgE by the immune system. Nat. Rev. Immunol. 14, 247–259.
Wu, J.N., 2005. An Illustrated Chinese Materia Medica. Oxford University Press, New York, pp. 262–263.
Zhao, Q., Lee, F.S., 1999. Mitogen-activated protein kinase/ERK kinase kinases 2 and 3 activate nuclear factor-κB through IκB kinase-α and IκB kinase-β. J. Biol. Chem. 274, 8355–8358.